KIRUBIA A

Data Scientist | ML Engineer | Analytics & AI

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PROFESSIONAL SUMMARY

Results-driven Data Scientist and ML Engineer with a B.Tech in AI & Data Science and hands-on experience building end-to-end machine learning pipelines, developing interactive dashboards, and deploying predictive models. Proficient in Python, SQL, Scikit-learn, XGBoost, and cloud-ready tools. Experienced with feature engineering, statistical analysis, NLP preprocessing, and model explainability (XAI). Adept at translating complex data insights into business-ready solutions using Power BI, Tableau, and Streamlit. Seeking a data science or ML engineering role to drive analytical impact at scale.

TECHNICAL SKILLS

ML / AI	Scikit-learn, XGBoost, LightGBM, Random Forest, KNN, SVM, Logistic Regression, K-Means, DBSCAN, PCA, SHAP (XAI), SMOTE, GridSearchCV, Cross-Validation
NLP & Gen AI	TF-IDF, tokenization, spaCy, NLTK, Hugging Face Transformers, LangChain basics, Prompt Engineering, RAG pipeline basics
Languages	Python, SQL (MySQL — joins, aggregations, subqueries, window functions, query optimization)
Data & EDA	Pandas, NumPy, SciPy, feature engineering, data cleaning, statistical analysis, hypothesis testing, outlier detection, correlation analysis
Visualization	Matplotlib, Seaborn, Plotly, Power BI (DAX basics), Tableau, Excel (Pivot Tables, Power Query, XLOOKUP)
Cloud & MLOps	GCP (BigQuery, Cloud Storage, Colab), MLflow (experiment tracking, model registry), FastAPI, Docker basics, GitHub, CI/CD basics, Postman
Deploy / Tools	Streamlit, Jupyter Notebook, Pickle/Joblib, VS Code, Google Colab, REST API integration

PROFESSIONAL EXPERIENCE

Data Science Intern — Gradtwin Services
Sept 2024 – Nov 2024

- Performed exploratory data analysis (EDA) on real-world datasets uncovering trends, patterns, and business-relevant outliers
- Built, tuned, and benchmarked supervised ML models (KNN, Random Forest) using cross-validation and GridSearchCV for hyperparameter optimization
- Engineered features via encoding (label, one-hot), normalization, and transformation pipelines to improve model performance
- Evaluated models using accuracy, RMSE, precision, recall, F1-score, and ROC-AUC; selected best approach through systematic comparison

PROJECTS

Global AI Adoption & Workforce Impact
XGBoost · LightGBM · K-Means · SHAP · MLflow · Streamlit

- Engineered full ML pipeline on 150,000+ company records to analyze AI adoption patterns across 20+ industries and global regions
- Built XGBoost multi-class classifier to predict AI adoption stage; applied SHAP for feature importance and model explainability
- Tracked all experiments using MLflow — logged parameters, metrics, and model artifacts for full reproducibility
- Applied K-Means clustering (elbow method) to segment companies into distinct AI maturity profiles for strategic insights
- Deployed interactive Streamlit dashboard with real-time filters and drill-down visualizations; handled class imbalance via SMOTE resampling

Sentiment Analysis — Product Reviews
Python · NLTK · spaCy · TF-IDF · Logistic Regression · Scikit-learn

- Built end-to-end NLP pipeline: text cleaning, tokenization, stopword removal, and TF-IDF vectorization for sentiment classification
- Trained and evaluated Logistic Regression classifier on public review dataset; visualized predictive features using word clouds and bar plots
- Applied class balancing and evaluated model using precision, recall, F1-score, and confusion matrix across positive and negative classes

India Housing Price Analysis
Python · Pandas · Seaborn · Streamlit · Regression

- Cleaned and analyzed large housing dataset; resolved missing values, duplicates, and outliers across multiple city-level records
- Conducted in-depth EDA using correlation heatmaps, pair plots, and statistical tests — identified location and property type as strongest pricing predictors
- Deployed Streamlit dashboard enabling users to explore regional pricing trends with dynamic filters

Sales Data Analysis & KPI Dashboard
Python · Tableau · Excel · SQL

- Analyzed 9,800+ sales records to evaluate revenue, profitability, and category-level trends using Python and SQL
- Developed Tableau dashboards for KPI tracking, discount impact analysis, and executive reporting; automated monthly reports via Excel Power Query
- Identified excessive discounting in select categories as a net revenue drain, delivering targeted corrective recommendations

Cirrhosis Disease Stage Prediction
Python · Scikit-learn · SMOTE · Random Forest · Logistic Regression

- Built multi-class classifier (Random Forest, Logistic Regression) on clinical patient data to predict cirrhosis disease stages
- Applied SMOTE and oversampling to address class imbalance; evaluated with accuracy, F1-score, and AUC-ROC across all stages

Bike Sales Pricing Analysis
Python · Excel · Matplotlib · Seaborn

- Analyzed pricing factors including mileage, age, brand, and location; generated visualizations to highlight market-level pricing patterns and support strategy

EDUCATION

B.Tech – Artificial Intelligence & Data Science | CGPA: 8.2
2021 – 2025

SIMATS School of Engineering, Chennai | Coursework: Machine Learning, Deep Learning, NLP, Statistics, Database Systems, Data Visualization